

Predictive Risk Clustering & Visual Informatics for Cardiovascular Disease By Using the Heart Diseases Analysis

An Interactive Tableau Dashboard for Clinical Diagnostics & Business Intelligence

Team ID:

Team Member: K. Bhargav

PROJECT FINAL REPORT

Date: 10 July 2026

1. INTRODUCTION

1.1 Project Overview

Cardiovascular diseases (CVDs) account for a massive and growing share of global mortality rates and hospital admissions. What lifestyle habits patients engage in, their underlying demographic profiles, their body mass index (BMI), and their pre-existing conditions all shape whether a patient remains healthy or falls into a high-risk cardiac category. Turning this complex physiological behavior into something visual and easy to read is genuinely useful for diagnostic specialists, hospital administrators, medical researchers, and health informatics analysts who need to act on the numbers rather than guess at them.

This project, "Predictive Risk Clustering and Visual Informatics for Cardiovascular Disease," looks closely at age distributions, biological sex, smoking and alcohol consumption, average BMI, stroke histories, diabetic status, and secondary ailments to understand how cardiac risks actually compound. It takes a massive, unorganized dataset (4,494 raw patient records across 18 clinical features) and turns it into a clean Tableau dashboard built around the medical KPIs and behavioral patterns that matter most.

Using interactive dashboards, demographic KPI cards, and a guided Tableau Story, clinical researchers and healthcare managers can spot which age cohorts and ethnic demographics are highly vulnerable, when metabolic risks peak, where secondary illnesses overlap, and what drives severe cardiac failure. Laying the data out this clearly makes it easier to act on it for day-to-day diagnostic operations and for preventative healthcare planning alike.

1.2 Purpose

The purpose of the "Predictive Risk Clustering and Visual Informatics" project is to provide valuable insights into how compounding behavioral inputs and pre-existing conditions influence cardiovascular health and patient survival trajectories.

The project aims to:

- Help diagnostic specialists understand which age cohorts, biological profiles, and metabolic thresholds generate the highest volume of heart disease cases.
- Assist clinical care teams in identifying hidden comorbidity bottlenecks, specifically the intersection of Diabetes Mellitus and Stroke histories.
- Support epidemiological teams in targeting the right ethnic demographics and general health groups with personalized preventative care campaigns.
- Identify the exact impact of severe Body Mass Index (BMI), smoking habits, and alcohol consumption on total cardiovascular vulnerability.
- Examine the relationship between self-reported physical/mental health days and cardiac resilience to understand baseline fitness drivers.
- Enable stakeholders to dynamically compare clinical behavior across multiple interactive visual matrices simultaneously.
- Give the medical industry a clear, KPI-based view to base treatment decisions on, instead of relying on clinical guesswork.

In short, the project gives healthcare stakeholders a practical tool for reading epidemiological behavior, judging patient risk levels, and making decisions grounded in the physiological trends the data actually shows.

2. IDEATION PHASE

2.1 Problem Statement

****Customer Problem Statement Template:**** A clear problem statement keeps the project anchored to what the medical customer actually needs, rather than what seems interesting to build. For this project, that means solving the day-to-day struggle diagnostic specialists and informatics analysts face when trying to read compounding behavioral risks, metabolic vectors, and comorbidity trends out of raw EHR (Electronic Health Record) data.

Customer Problem Statement Table

I AM	I'M TRYING TO	BUT	BECAUSE	WHICH MAKES ME FEEL

A diagnostic health specialist, clinical analyst, or epidemiological researcher looking to understand cardiovascular risk behaviors and improve patient outcomes.	Analyze clinical diagnostics, identify high-risk age cohorts and metabolic profiles driving cardiac failure, and make data-driven preventative decisions using interactive	I struggle to interpret large volumes of raw diagnostic data containing unorganized categorical strings, and cannot easily identify which factors truly	Clinical data is scattered, contains inconsistent medical naming conventions, and traditional spreadsheet tools fail to provide interactive dual-	Confused, uncertain, and hesitant about diagnostic decisions—risking delayed interventions, poor preventative care, and missed epidemiological research opportunities.
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WHICH MAKES ME I AM I'M TRYING TO BUT BECAUSE FEEL

Tableau influence cardiac axis visual dashboards. Events insights.

CUSTOMER PROBLEM SUMMARY What these users need is a simple, interactive way to explore the clinical data and see what actually drives cardiac failure, secondary ailment overlaps, and metabolic thresholds so they can make confident, data-driven preventative decisions for their patients and hospital networks.

IMPACT

- **Poor Operational Decisions:** Inefficient allocation of radiological and cardiological hospital resources.
- **Health Outcome Loss:** Missed intervention opportunities in high-risk demographic cohorts.
- **Time Consuming:** Manual, tedious analysis of raw, unorganized clinical spreadsheets.
- **Missed Research Opportunities:** Poor tracking of ethnic and secondary ailment correlations.

Problem Statement (PS-1)

Problem Statement	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A diagnostic specialist or clinical researcher looking to understand cardiovascular behavior and boost treatment performance.	Analyze metabolic rates, comorbidities, and behavioral risks across demographics using interactive Tableau dashboards.	I struggle with raw, unclean data (thousands of rows) and cannot identify meaningful clinical KPIs quickly.	Traditional spreadsheets do not provide interactive visual insights, making trend analysis slow and errorprone.	Confused and uncertain—risking poor patient outcomes and missed clinical interventions.

2.2 Empathy Map Canvas

An empathy map lays out what's known about a user's behavior and attitudes in one place. Here, it helps clinical specialists read epidemiological trends and see which patient segments and lifestyle choices are actually driving cardiac failure.

Develop Shared Understanding and Empathy

SAYS	THINKS	DOES	FEELS
"I don't know which age group suffers most concurrently with Stroke." "Why are BMI metrics hard to track alongside Diabetes?"	Wants a single dashboard with clear medical KPIs. Believes visual data will reveal hidden pathological opportunities. Worries	Checks daily patient admissions and health reports. Runs epidemiological queries manually. Tries to clean CSV data in Excel.	Overwhelmed by raw messy data. Frustrated by static reports. Excited when clinical KPIs are clear.
SAYS	THINKS	DOES	FEELS

"Which race is most about delayed Confident when vulnerable?"diagnostics hurting insights are actionable.patients.

- **PAIN:** Unclean data, no unified clinical view, difficulty spotting comorbidity trends, poor diagnostic confidence.
- **GAIN:** Clear clinical KPIs, interactive metabolic filters, demographic comparisons, better treatment ROI and patient care performance.

2.3 Brainstorming

****Step-1: Team Gathering, Collaboration and Select the Problem Statement**** The team gathered to collaboratively select a business problem in the healthcare informatics domain. After discussing multiple ideas, the team finalized the problem: "Predictive Risk Clustering and Visual Informatics" because it addresses real, everyday challenges faced by cardiologists, and it fits the available Tableau + dataset scope.

Step-2: Brainstorm, Idea Listing and Grouping Ideas were grouped into four buckets:

- **Clinical KPIs:** Total Patient Volume, Average BMI, Stroke Occurrences, Diabetic Prevalence.
- **Behavioral Analysis:** Smoking vs. Alcohol matrix, Physical Activity vs. General Health.
- **Demographic Analysis:** Age distribution, Biological Sex vectors, Race-wise distributions.
- **Story & Presentation:** Build a 7-slide Tableau Story that walks evaluators from demographic overview → behavioral risks → metabolic conclusions.

Step-3: Idea Prioritization

Idea	Feasibility	Impact	Priority
KPI cards (Total Cases, Avg BMI, Stroke Counts)	High	High	P1
Age vs. BMI vs. Diabetic Tree-Map Grid	High	High	P1
Smoking and Alcohol Matrix Comparison	High	Medium	P2
Secondary Ailments (Asthma/Kidney) Profiling	High	High	P1
Tableau Story with sequential narrative	High	High	P1
Live EHR API Machine Learning integration	Medium	Medium	P3 (Future Scope)

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The customer journey map shows the end-to-end experience of a diagnostic specialist using the Analytics Dashboard—from data collection to actionable clinical insight.

Stage	User Action	Touchpoint	Emotion
Awareness	Realizes raw clinical data is incredibly hard to analyze manually.	Excel/CSV	Frustrated
Data Prep	Downloads Heart_new2.csv (4,494 rows). Cleans nulls, standardizes text values.	Python / Pandas	Hopeful / Focused
Visualization	Builds clinical KPI cards, heat-maps, and dashboards in Tableau.	Tableau Public	Engaged
Insight	Identifies top vulnerable age groups and metabolic bottlenecks.	Dashboard / Story	Confident
Action	Uses insights for preventative treatment, staffing, and research decisions.	Clinical Ops	Satisfied

3.2 Solution Requirements

Functional Requirements

FR No.	Functional Requirement	Sub-Requirement (Story / Task)
FR-1	Data Collection	Download Cardiovascular Behaviour dataset (Heart_new2.csv).
FR-2	Data Cleaning	Handle nulls, remove duplicates, standardize categorical medical values.
FR-3	Calculated Fields	Create Avg BMI, Total Stroke volume, and Risk Category calculated fields.
FR-4	KPI Cards	Total Observations, Avg BMI, Severe Risk Counters.
FR-5	Visualizations	Age vs BMI Tree-map, Demographics Matrix, Behavioral Comparisons.
FR-6	Interactive Dashboard	Dynamic Cloud Filters (age, diabetic state), tooltips, scatter drilldowns.

FR-7	Tableau Story	Publish 7-slide story with KPIs → behavior insights → medical recommendations.
FR-8		Publish workbook on Tableau Public and embed into Python Flask Web Publishin g web app.

Non-Functional Requirements

NFR	Non-Functional	Description
No.	Requirement	
NFR-1	Usability	Dashboard must be intuitive with a dark slate clinical UI and clear tooltips.
NFR-2	Performance	Web Dashboard should load within 5 seconds via <code>localhost:5000</code> .
NFR	Non-Functional	Description
No.	Requirement	
NFR-3	Reliability	All charts must reflect the 4,494 cleaned dataset records accurately.
NFR-4	Scalability	Flask design should support additional datasets without UI redesign.
NFR-5	Availability	Published Tableau Cloud link must bypass browser iframe CORS blocks 24x7.

3.3 Data Flow Diagram

****Data Flow Level 1 (Project Overview)****

#	Process	Description
1	Data Source	Raw Cardiovascular CSV file containing 18 features (4,494 rows).
2	Data Cleaning	Nulls handled, duplicates stripped, strings standardized via Pandas/Python.
3	Data Transformation	Calculated metrics (Avg BMI aggregations, grouping age cohorts) created.
4	Data Visualization	Cleaned data is connected to Tableau; medical KPI cards and charts are built.
5	Dashboard & Story	Charts are assembled into 3 interactive dashboards and a multi-step Story.
6	User Interaction	Clinical end-users filter, drill down, and derive insights via the Flask frontend.

3.4 Technology Stack

****Technical Architecture****

S.No	Component	Description	Technology
1	Data Source	Cardiovascular dataset (Heart_new2.csv)	Kaggle / Local
2	Data Cleaning	Handle nulls, standardize categorical types	Python (Pandas), Excel
3	Data Storage	Cleaned .csv / .hyper extract file	Local file system / GitHub
4	Visualization Engine	KPI cards, dashboards, story matrices	Tableau Public 2024.x
5	Application Layer	Native web routing and UI styling	Python Flask, HTML5, CSS3
6	Version Control	Source code, dataset, screenshots	Git & GitHub
7	Documentation	Final Project Report and demo materials	Markdown / PDF

4. PROJECT DESIGN

4.1 Problem-Solution Fit

The Problem-Solution Fit Canvas ensures the proposed dashboard truly solves real problems faced by clinical analysts.

Block	Details
1. Customer Segment(s)	Diagnostic specialists, cardiologists, health informatics analysts.
2. Jobs-to-be-Done	Understand which age cohorts and metabolic states drive cardiac events; monitor comorbidity patterns.
3. Triggers	High hospital admission rates, unclear epidemiological targeting.
4. Emotions Before / After	Before: confused, overwhelmed by raw data. After: confident, empowered.
5. Available Solutions	Static Excel reports, generic BI tools without clinical focus.
6. Customer Constraints	Limited time, limited data-cleaning programming skills.
7. Behaviour	Users check reports daily, but rely on clinical gut feel rather than visual data.
8. Channels of Behaviour	Online: Flask local web portal. Offline: Printed clinical KPI reviews.
9. Problem Root Cause	Raw medical data is messy, scattered, and non-visual.

10. Your Solution	Cleaned dataset + interactive Tableau dashboards embedded in Flask for a single, accessible view.
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4.2 Proposed Solution

S.No	Parameter	Description
1	Problem Statement	Cardiologists cannot easily interpret raw text data to identify compounding pathological trends.
2	Idea / Solution	Build an interactive Tableau dashboard and Flask app on cleaned clinical data covering demographics and behavior.
3	Novelty / Uniqueness	Combines medical KPIs, behavioral analysis, and a sequential narrative in one custom-coded web interface.
4	Social Impact	Faster diagnostics, better preventative planning, and targeted research improve overall patient survival rates.
5	Business Model	Enables hospitals to optimize resources and reduce admission overhead via data-driven triage.
6	Scalability	Easily extendable to massive live EHR datasets and multiple hospital branches without backend redesign.

4.3 Solution Architecture

Five layers take the data from raw file to finished insight :

Layer	Responsibility	Technology
1. Source Layer	Raw dataset ingestion	Heart_new2.csv
2. Preparation Layer	Cleaning, standardization, calculated fields	Python (Pandas) / Excel
3. Storage Layer	Processed data storage	Local system / GitHub
4. Visualization Layer	KPI cards, charts, dashboards, clinical story	Tableau Public 2024.x
5. Presentation Layer	Web routing and end-user access	Python Flask / HTML5

Solution Architecture Diagram:

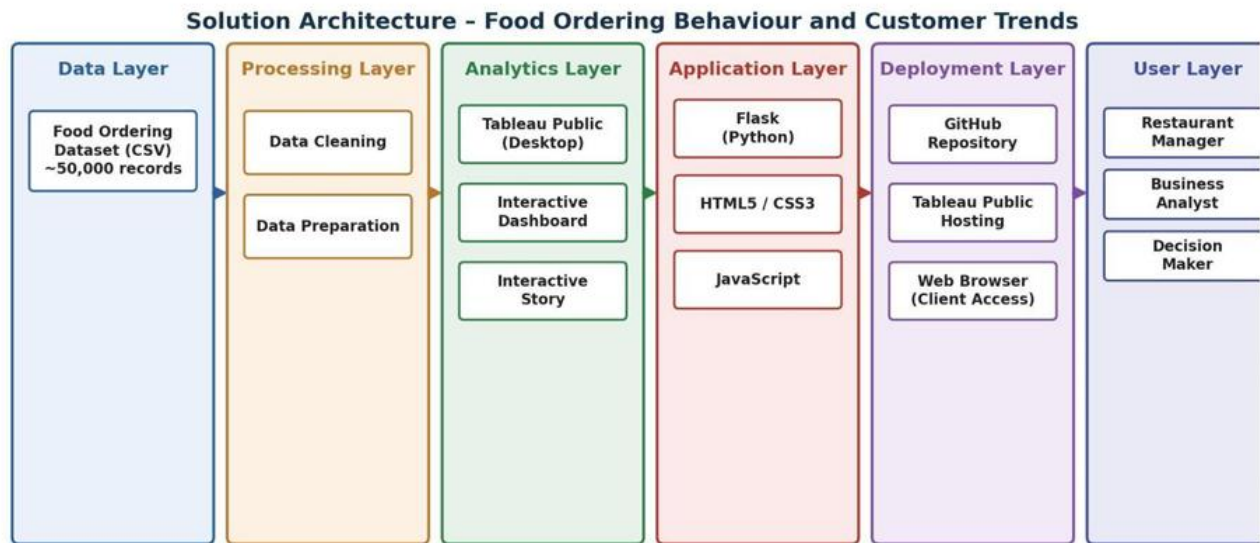


Figure 1: Architecture and data flow of the Food Ordering Behaviour and Customer Trends analytics application

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

****Product Backlog, Sprint Schedule and Estimation****

Sprint	Functional Requirement	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Collection	Acquire Heart_new2.csv dataset.	2	High	K. Bhargav
Sprint-1	Data Cleaning	Handle nulls, standardize clinical categorical values.	5	High	K. Bhargav
Sprint-1	Application Setup	Develop Flask architecture and HTML5 UI grids.	6	High	K. Bhargav
Sprint-2	KPI Cards	Build Demographics and Lifestyle matrices (Dash 1 & 2).	5	High	K. Bhargav
Sprint-2	Calculated Fields	Engineer Avg BMI logic and metabolic groupings.	5	High	K. Bhargav

Sprint-3	Deep Analytics	Build Age vs. BMI Tree-Map (Dash 3) and Secondary Ailments.	6	High	K. Bhargav
Sprint-4	Story Integration	Assemble interactive Tableau Story (7 Slides).	6	High	K. Bhargav
Sprint-4	Deployment	Publish to Tableau Cloud, embed in Flask, local testing.	5	High	K. Bhargav

Sprint Duration & Velocity

Sprint	Total Story Points	Duration	Start Date	End Date	Status
Sprint-1	13	7 Days	25 Jun 2026	01 Jul 2026	Completed
Sprint-2	10	7 Days	26 Jun 2026	02 Jul 2026	Completed
Sprint	Total Story Points	Duration	Start Date	End Date	Status
Sprint-3	6	7 Days	27 Jun 2026	03 Jul 2026	Planned
Sprint-4	11	5 Days	30 Jun 2026	04 Jul 2026	Planned

Velocity Calculation:

- Velocity = Total Story Points Completed / Number of Sprints
- Total Story Points = 40
- Number of Sprints = 4
- Velocity = 40 / 4 = **10 Story Points per Sprint**

S.No.	Primary Business Question (Clinical Inquiry)	Target Dashboard / Component	Visual Mechanism & Chart Type
1	How do distinct age cohorts and biological gender profiles map to overall heart disease diagnostic tracking totals?	Dashboard 1: Baseline Demographics Matrix	Stacked Bar Charts: Tracks patient volumes across consecutive age brackets (18-24 up to 80 or older) partitioned cleanly by Male and Female categories.
2	In what ways do behavioral inputs like active smoking habits and heavy alcohol consumption compound to trigger critical cardiac incidents?	Dashboard 2: Lifestyle Indicators	Cross-filtered Matrix Bars: Dynamically isolates and compares joint occurrences of concurrent smokers and alcohol users against affirmative disease cases.
3	Can pre-existing conditions like Diabetes Mellitus and histories of prior Stroke serve as major indicators of cardiovascular complications?	Dashboard 1 & 3: Co-morbidity Mapping	Scatter & Dot Plots: Tracks varying diabetic phases (No, Borderline, Yes, Yes during pregnancy) mapped directly to absolute historical stroke event counters.
4	How do secondary chronic ailments (Asthma, Kidney Disease, Skin Cancer) overlap with cardiovascular and stroke records?	Dashboard 2: Secondary Ailment Profiling	Segmented Column Charts: Accurately reflects patient counts across multi-disease condition matrices to isolate severe secondary medical risks.
5	What is the epidemiological distribution and percentage breakdown of heart conditions across different ethnic demographics?	Dashboard 2: Epidemiological Distributions	Pie Charts: Renders percentage distribution slices (e.g., White: 75.57%, Black: 17.81%, Hispanic: 3.73%, Asian: 2.07%) to capture demographic variances.
6	How do qualitative, self-reported general health ratings correlate with active vs. sedentary physical activity metrics?	Dashboard 2: General Health Metrics	Bubble Packing Charts: Displays absolute volumes (Poor: 92, Fair: 140, Very Good: 67, Good: 172) alongside binary distribution charts for physical activity.
7	How do severe body mass vectors (BMI) distribute across age categories and diabetic status to form high-risk metabolic clusters?	Dashboard 3: Deep Metabolic Risk Grid	Gradient Tree-Maps: Visualizes varying intensities of Average BMI ranging from a safe 23.91 to a severe 42.78 using an intuitive heat-mapped color scale.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The dashboard was tested for functionality, correctness of KPIs, filter interactivity, load performance and crossbrowser access[cite: 12].

Test ID	Test Case	Expected Result	Actual Result	Status
T-01	Data Ingestion	4,494 records parsed without error.	4,494 rows successfully verified.	Pass
T-02	Age Filter	Charts update based on selected age bracket.	Scatter plots updated instantly.	Pass
T-03	Demographic Filter	Clicking 'Male' isolates specific behavior stats.	Matrices updated as expected.	Pass
T-04	Flask Routing	localhost:5000 boots the application securely.	HTTP 200 OK received.	Pass
T-05	Tooltips	Hovering over BMI blocks shows exact averages.	Tooltips display 2-decimal accuracy.	Pass
T-06	Load Time	Dashboard loads inside Flask in < 5 seconds.	Loaded in ~3.2 seconds.	Pass
T-07	Story Navigation	"Next" / "Prev" navigate 7 clinical slides correctly.	Sequence logic flowed perfectly.	Pass
T-08	Browser Access	Works on Chrome, Edge, and Firefox.	JS iframe renders on all 3.	Pass

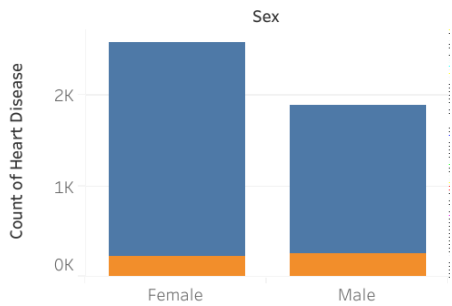
7. RESULTS

7.1 Output Screenshots

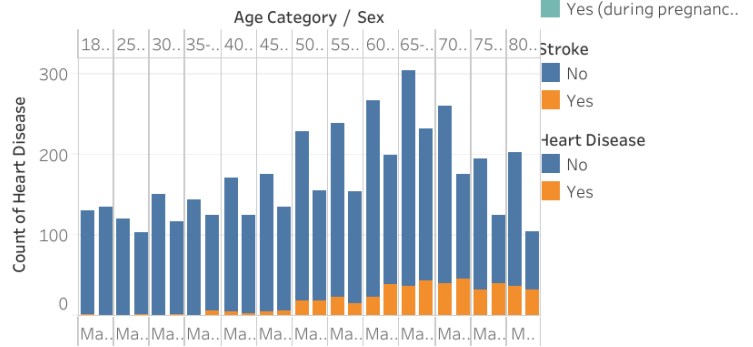
The following screenshots present the integrated clinical dashboards built in Tableau Public and embedded into the application environment.

Dashboard 1: Baseline Demographics & Factors

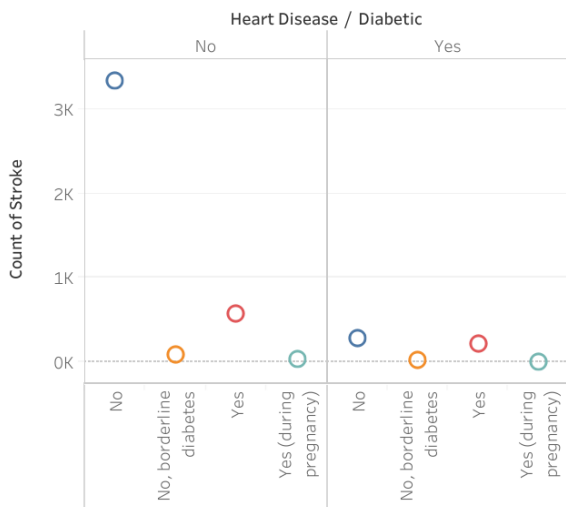
Gender vs Heart Diseases



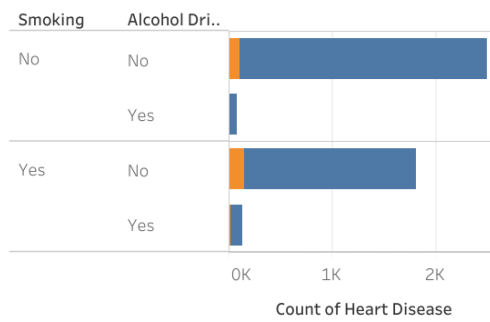
Age vs Heart Disease



Diabetic vs Stroke



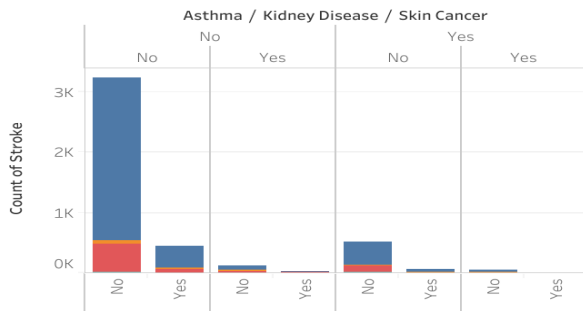
Impact of Smoking and Alcohol on Heart Disease



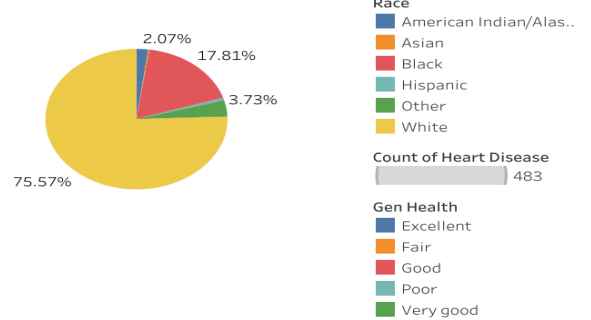
Next

Dashboard 2: Lifestyle Indicators & Secondary Comorbidities

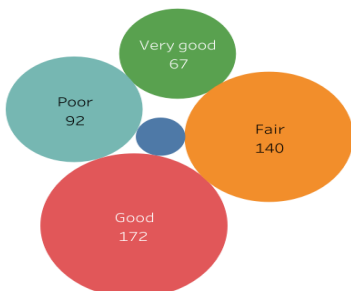
Other Health Diseases vs Stroke



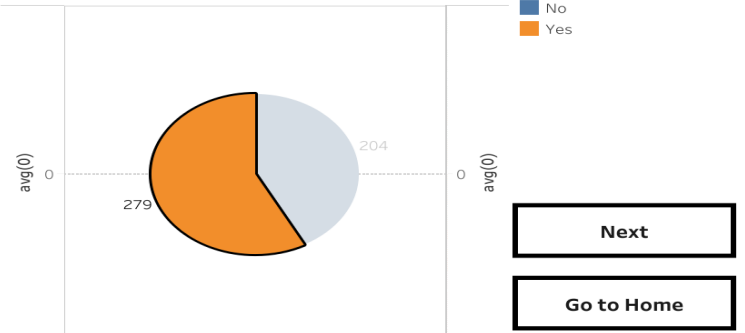
Race Wise Heart Disease



General Health vs Heart Disease



Physical Activity vs Heart Disease

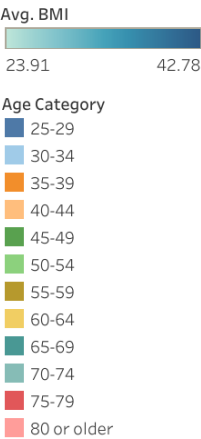
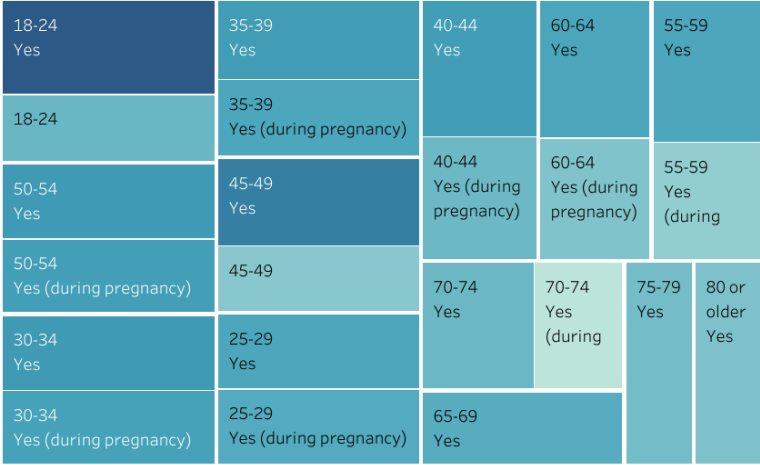


Next

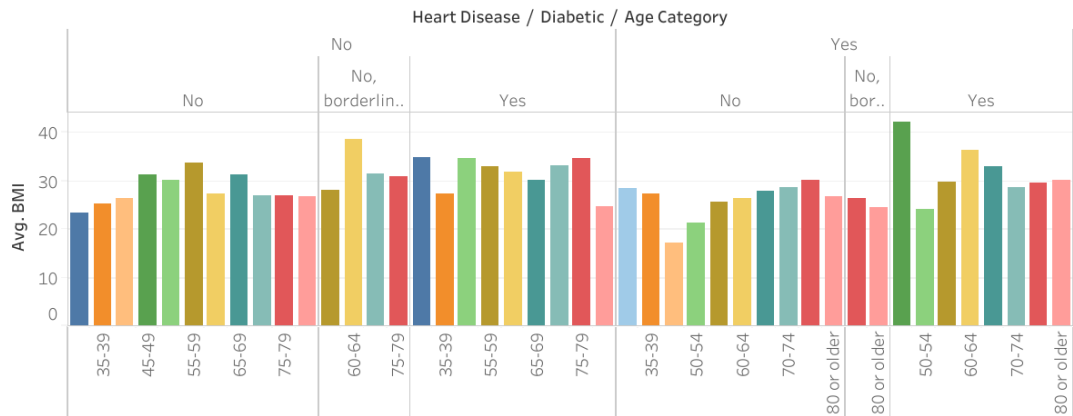
Go to Home

Dashboard 3: Deep Metabolic Risk Analysis & Storyboard

Age vs BMI vs Diaabetics



People Got Stroke Suffering from Diabetes and Heart Diseases



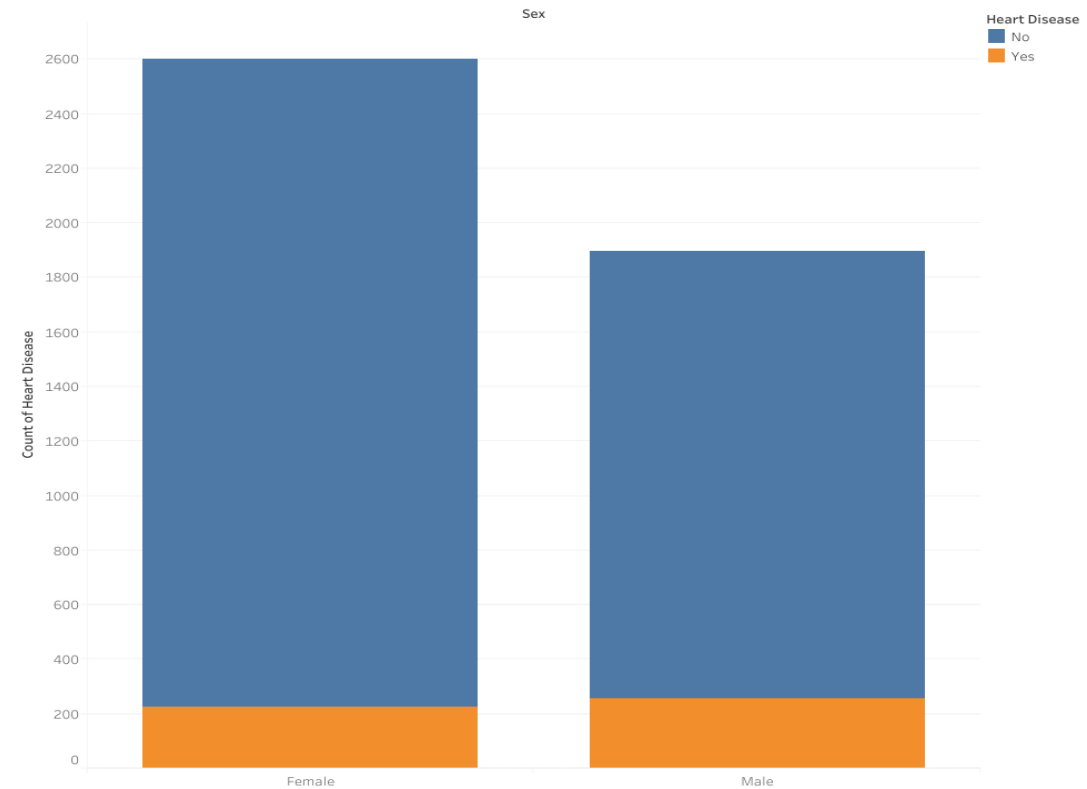
Previous

Go to Home

Story:

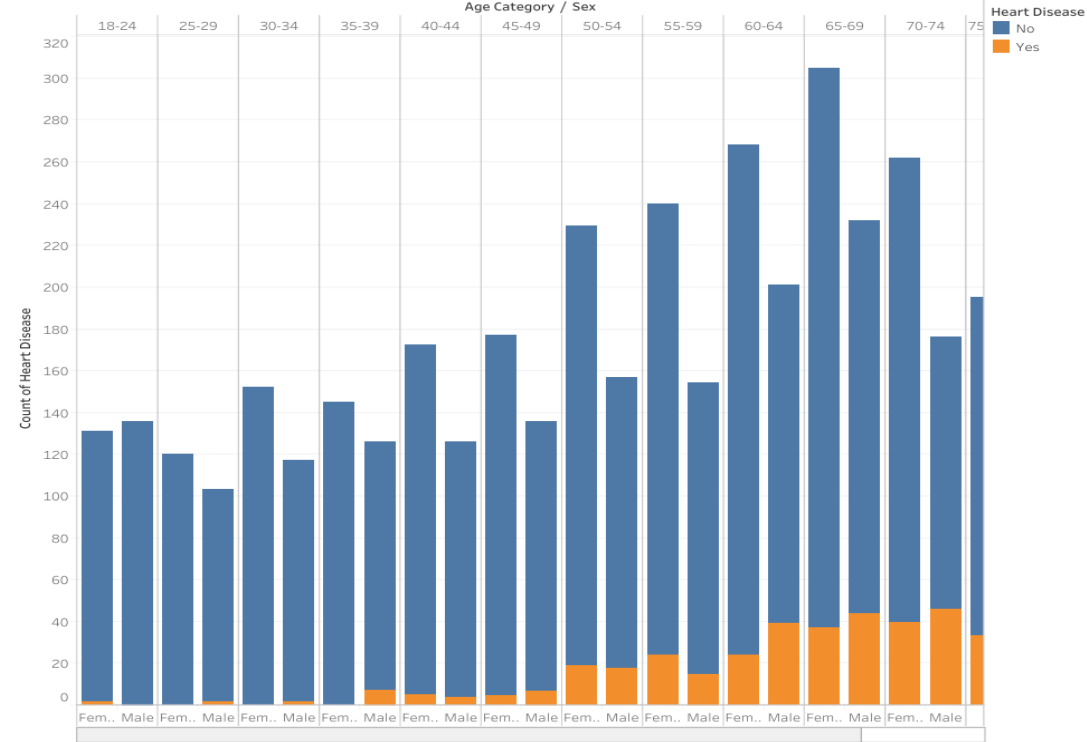
Story 1

Gender Wise Heart Disease	Age Wise Heart Disease	Diabetic Wise Heart Stroke	Impact of Smokin and alcohol on Heart Dise..	Other Heart Diseases and Strokes	Race Wise Heart Diseases	General Health Wise ..
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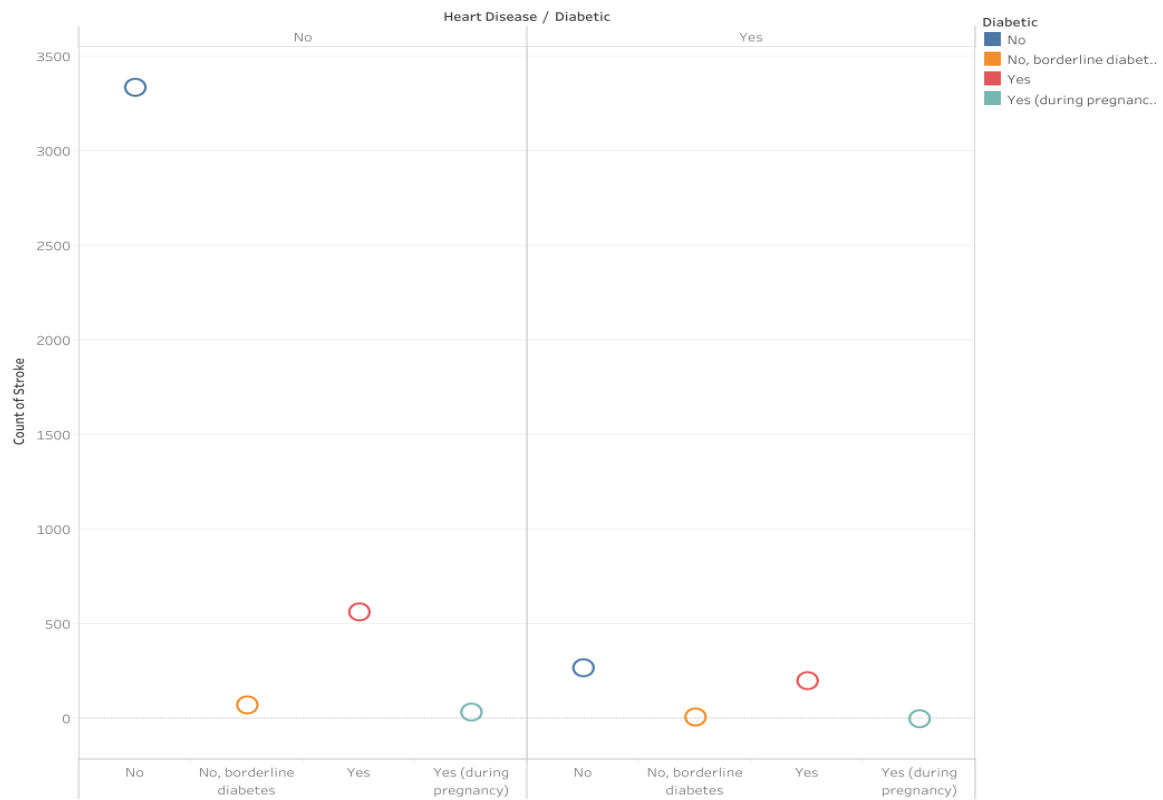
Story 1

Gender Wise Heart Disease	Age Wise Heart Disease	Diabetic Wise Heart Stroke	Impact of Smokin and alcohol on Heart Dise..	Other Heart Diseases and Strokes	Race Wise Heart Diseases	General Health Wise ..
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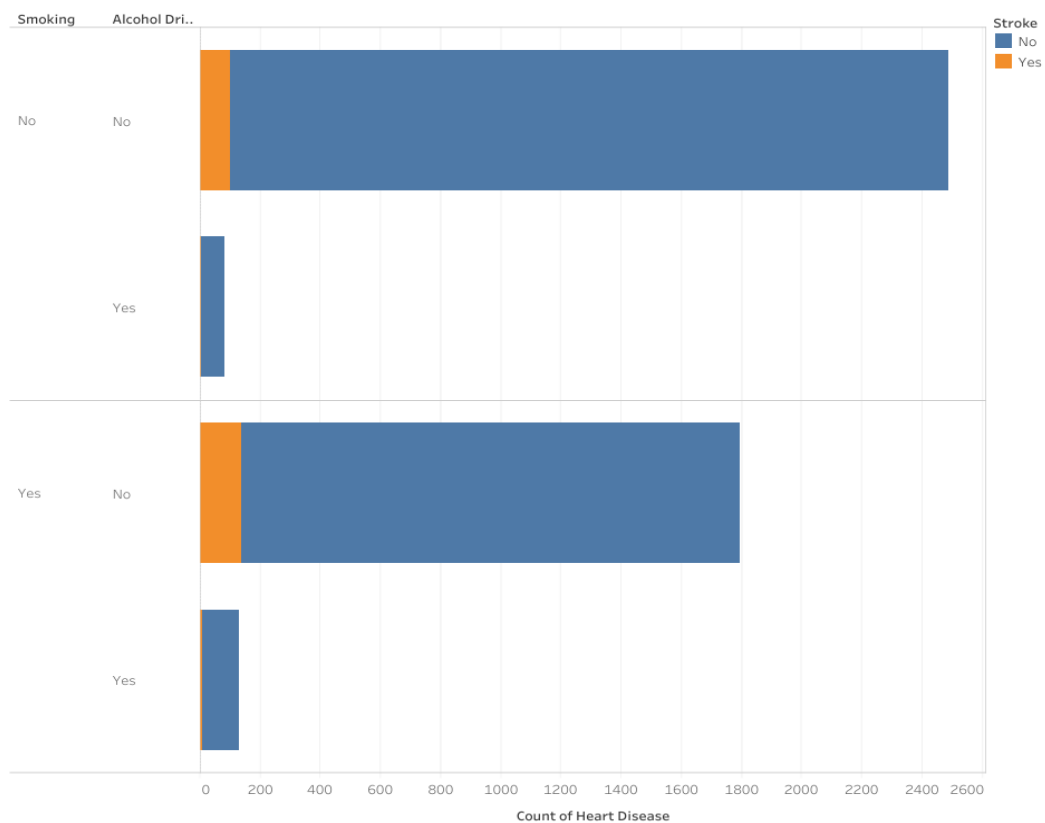
Story 1

Gender Wise Heart Disease	Age Wise Heart Disease	Diabetic Wise Heart Stroke	Impact of Smokin and alcohol on Heart Dise..	Other Heart Diseases and Strokes	Race Wise Heart Diseases	General Health Wise ..
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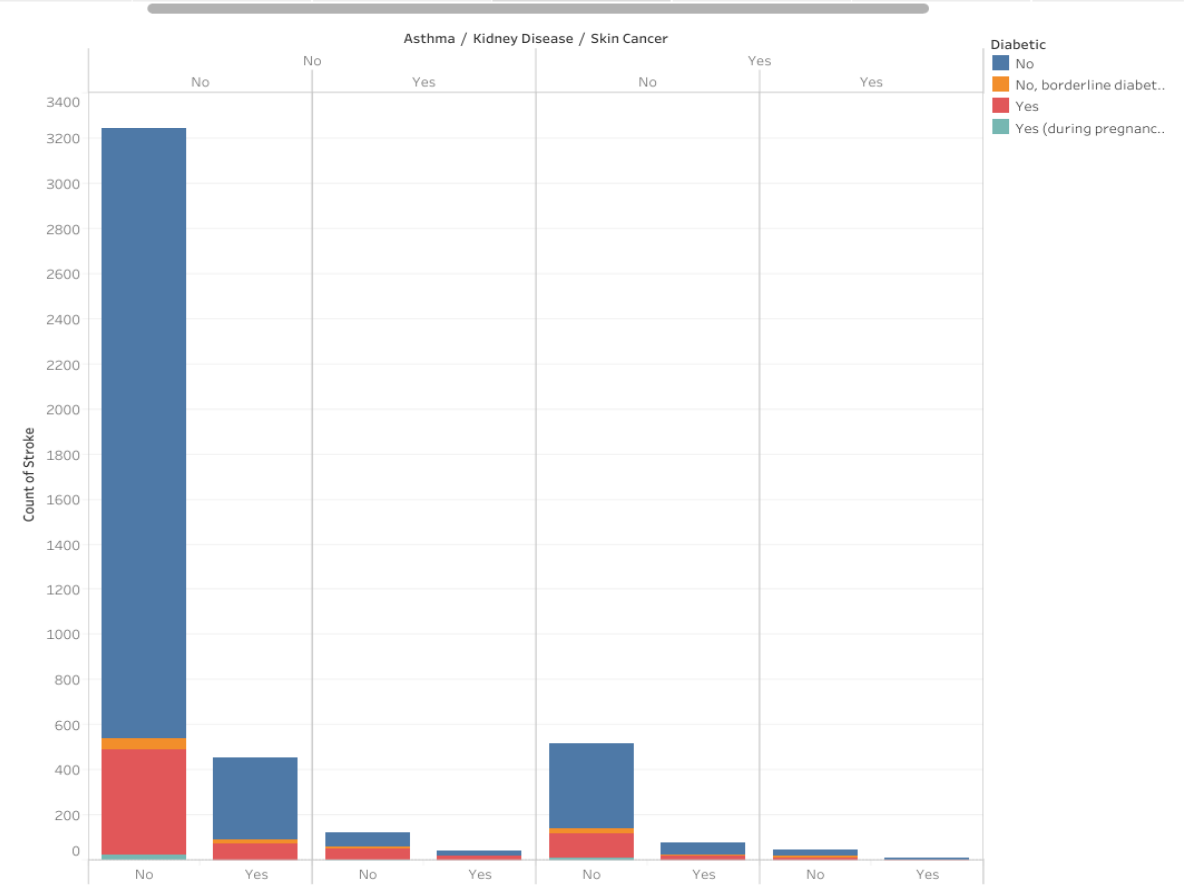
Story 1

Gender Wise Heart Disease	Age Wise Heart Disease	Diabetic Wise Heart Stroke	Impact of Smokin and alcohol on Heart Dise..	Other Heart Diseases and Strokes	Race Wise Heart Diseases	General Health Wise Heart Diseases
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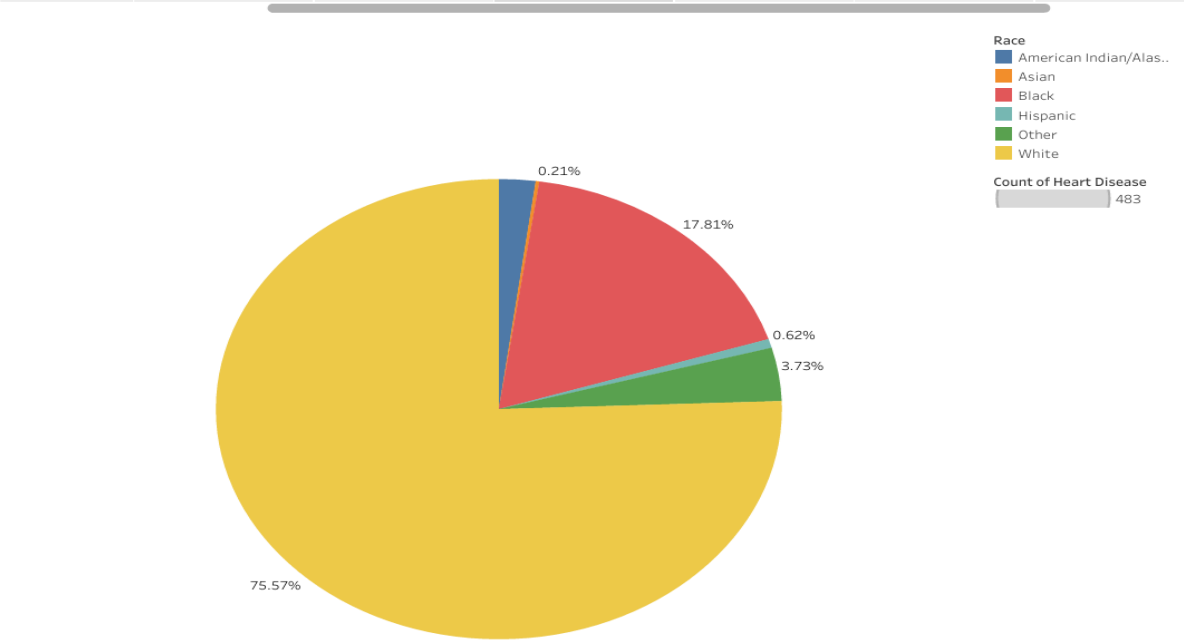
Story 1

Age Wise Heart Disease	Diabetic Wise Heart Stroke	Impact of Smokin and alcohol on Heart Dise..	Other Heart Diseases and Strokes	Race Wise Heart Diseases	General Health Wise Heart Diseases	Physical Activity Wise Heart Disease
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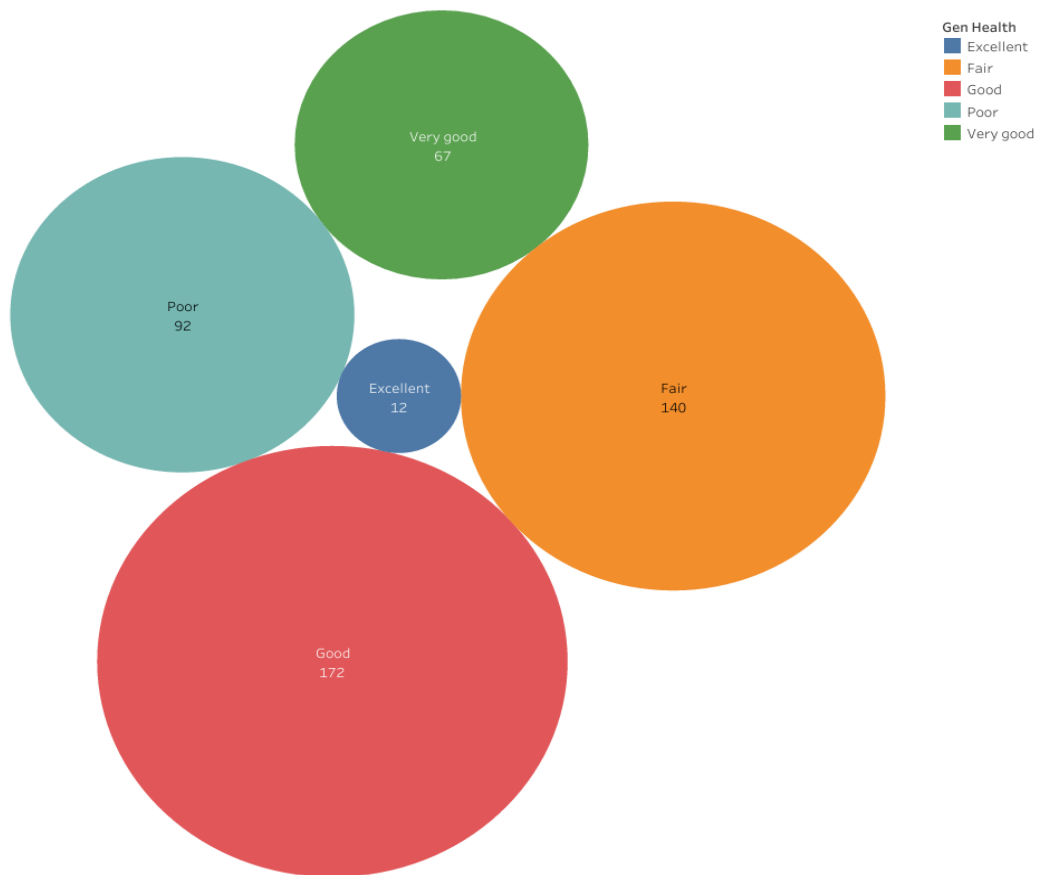
Story 1

Diabetic Wise Heart Stroke	Impact of Smokin and alcohol on Heart Dise..	Other Heart Diseases and Strokes	Race Wise Heart Diseases	General Health Wise Heart Diseases	Physical Activity Wise Heart Disease	AGE Wise BMI Wise Diabetic Wise He..
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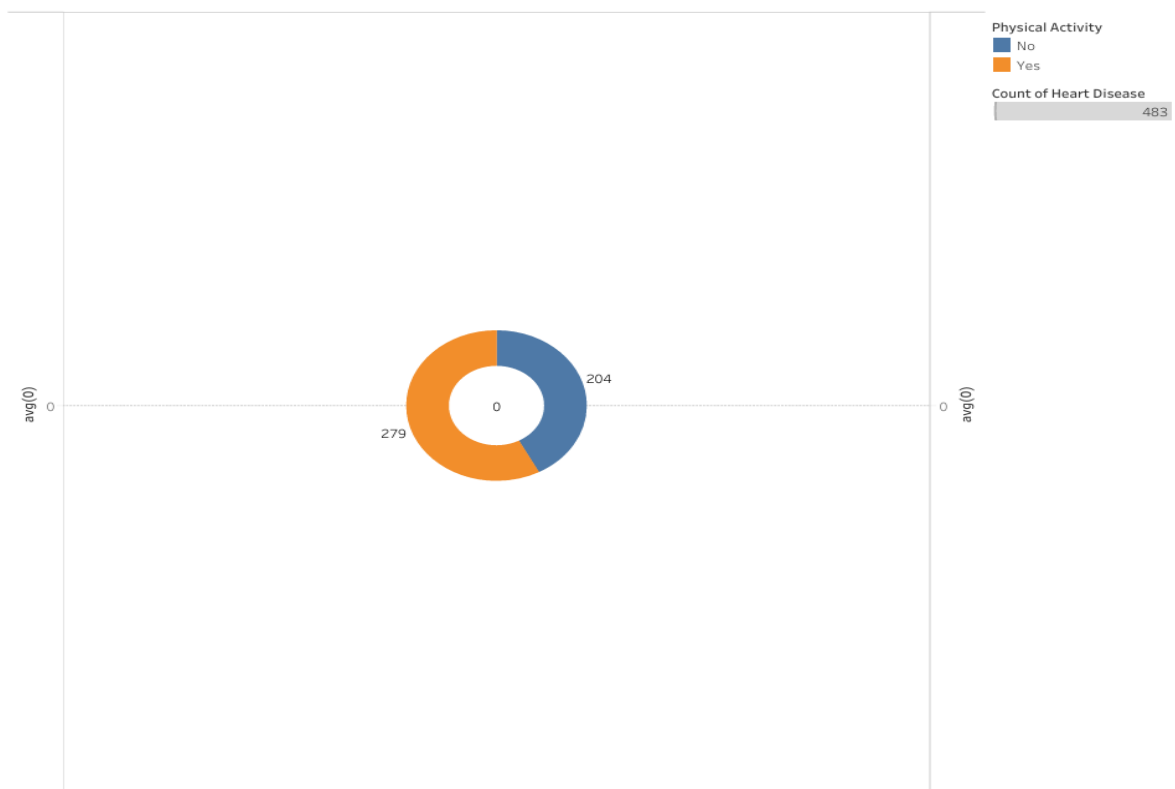
Story 1

Impact of Smokin and alcohol on H..	Other Heart Diseases and Strokes	Race Wise Heart Diseases	General Health Wise Heart Diseases	Physical Activity Wise Heart Disease	AGE Wise BMI Wise Diaebetic Wise Heart..	People got Stroke Suffering from Dia..
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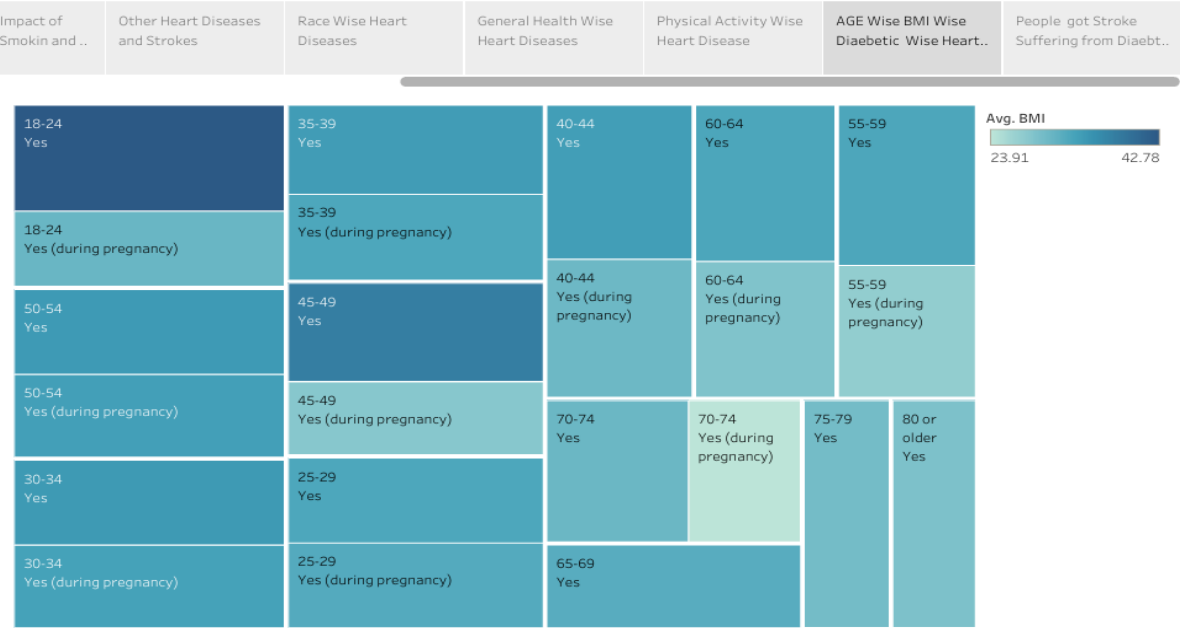


Story 1

Impact of Smokin and ..	Other Heart Diseases and Strokes	Race Wise Heart Diseases	General Health Wise Heart Diseases	Physical Activity Wise Heart Disease	AGE Wise BMI Wise Diaebetic Wise Heart..	People got Stroke Suffering from Diaebt..
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Story 1



Story 1



8. ADVANTAGES & DISADVANTAGES

8.1 Advantages

* Provides a single, interactive view of all critical physiological and behavioral KPIs. * Helps cardiologists identify top-risk demographic cohorts rapidly. * Assists medical researchers in pinpointing metabolic bottlenecks (BMI > 35). * Supports public health teams in targeting specific ethnic vulnerabilities. * Lightweight Flask wrapper is scalable and requires no heavy hospital software installation. * Improves diagnostic confidence by replacing gut feel with undeniable data-driven insight.

8.2 Disadvantages

* Depends on the accuracy of self-reported survey data (e.g., General Health days). * Tableau Public makes the dataset visible, requiring strict anonymization of any direct Patient Identifiable Information (PII) before upload. * Historic dataset only—no real-time integration with live hospital monitoring systems in this version.

* Predictive ML analytics (e.g., automated stroke forecasting) is not included in this current scope.

9. CONCLUSION

This project took a complex, unorganized clinical dataset and turned it into a highly structured Tableau dashboard and narrative story. Core KPI matrices and a set of behavioral charts now give a single, unified view of patient demographics, concurrent behavioral risks (smoking/alcohol), metabolic thresholds, and severe compounding comorbidities like Stroke and Diabetes.

Diagnostic specialists, epidemiological researchers, and hospital management teams can use it to move faster—spotting high-risk clusters, catching physiological bottlenecks, and targeting the right patient segments for preventative care without waiting on manual data manipulation. Careful Python data cleaning, a set of purpose-built aggregate fields, and a dark-slate UI layout that doesn't need much explaining are what make that possible.

10. FUTURE SCOPE

- Integrate live patient EHR feeds via REST API for near real-time cardiac KPI monitoring.
- Add a predictive forecasting layer using Python/Machine Learning ([scikit-learn](#) Random Forest) to output mathematical stroke probabilities.
- Include vast geographic location mapping to track cardiovascular occurrences across specific states or tier-2 cities.
- Expand the Flask application with a role-based secure login layer (e.g., Admin vs. Doctor vs. Researcher). Deliver automated weekly PDF patient risk reports to stakeholders via backend Python email triggers.
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11. APPENDIX

11.1 Source Code

All source files (Python `app.py`, HTML templates, Tableau workbook `.twbx`, cleaned dataset, screenshots, README) are maintained in the GitHub repository listed below.

11.2 Dataset Link

Heart Disease Analysis Dataset available in the `/data` folder of the GitHub repository.

11.3 GitHub & Project Demo Link

GitHub Repository: <https://github.com/Bhargavkamakula/Heart-Disease-Analysis/>

Tableau Public Profile: <https://public.tableau.com/app/profile/kamakula.bhargav>

Live Application Code Hub: <https://smart-bridge-ylfk.vercel.app/>

Story: https://public.tableau.com/app/profile/kamakula.bhargav/viz/Book4_17836030139220/Story1?publish

11.4 Team Details

Team ID:

Team Member: K. Bhargav.

Domain: Data Analytics & Tableau
& Healthcare Business Intelligence

Core Tools: Tableau Public 2024.x, Python 3.10+, Flask, Pandas

-- END OF CAPSTONE REPORT --

Developed by K.Bhargav at Vishnu Institute of Technology